

# **Supervised learning (II): Classification (II)**

- regularization for classification**
- support vector machines**

**BIOST 2155**

**Class 6**  
**October 2nd, 2026**

# Support vectors

## Introducing hyperplanes

Key idea: We try and find a plane that separates the classes in feature space

If we cannot, we get creative in two ways:

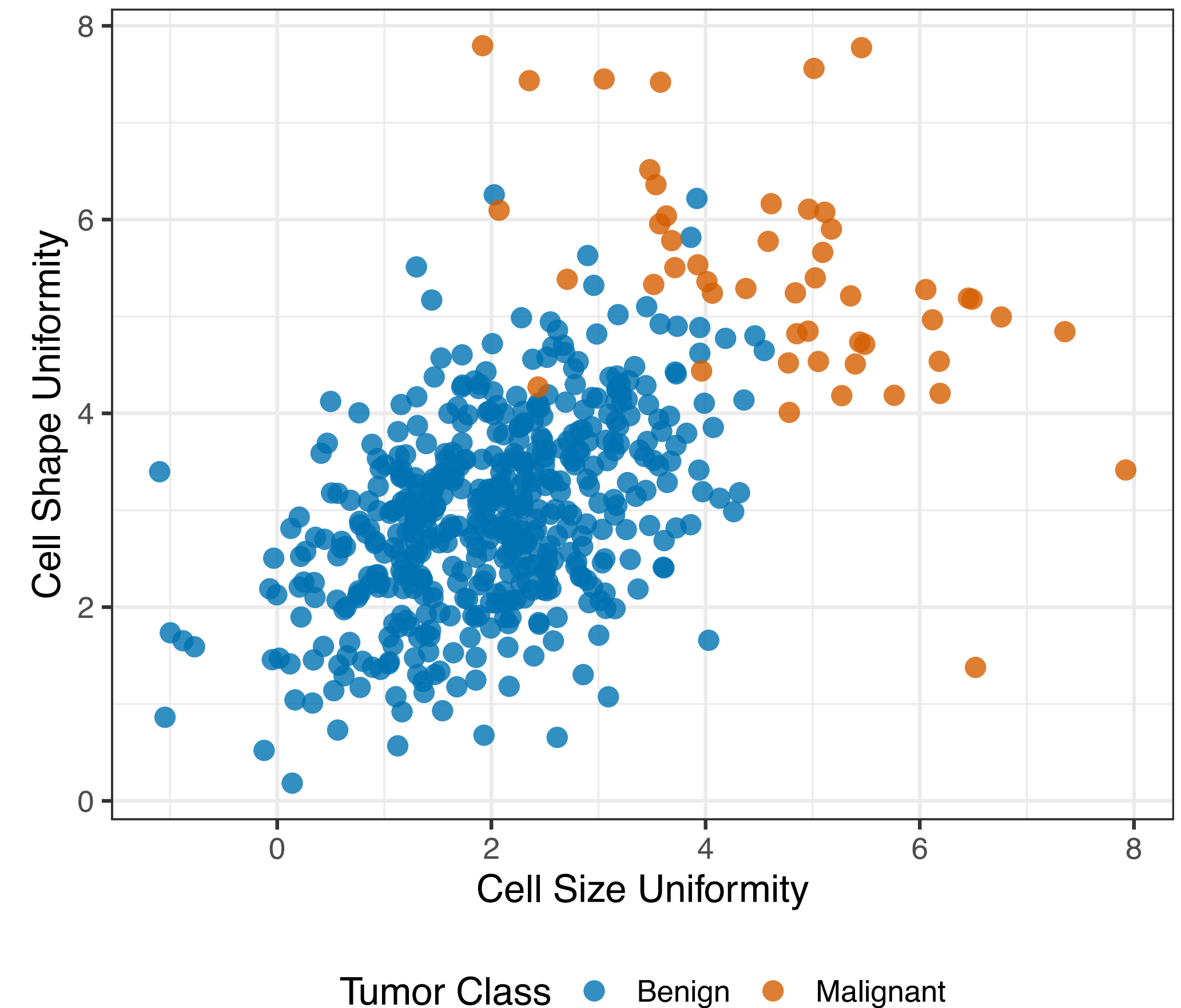
- We soften what we mean by “separates”, and
- We enrich and enlarge the feature space so that separation is possible.

Consider the following motivating example:  
Want to classify patient tumors as benign or malignant based on biomarkers.

Two features: Cell Size Uniformity and Cell Shape Uniformity.

### Tumor Classification based on Biomarkers

How would you draw a line to separate the classes?



We will focus on binary classification for now and extend to multi-class problems later

# What is a hyperplane?

## Some math + visualizing for $p = 2$ dimensions

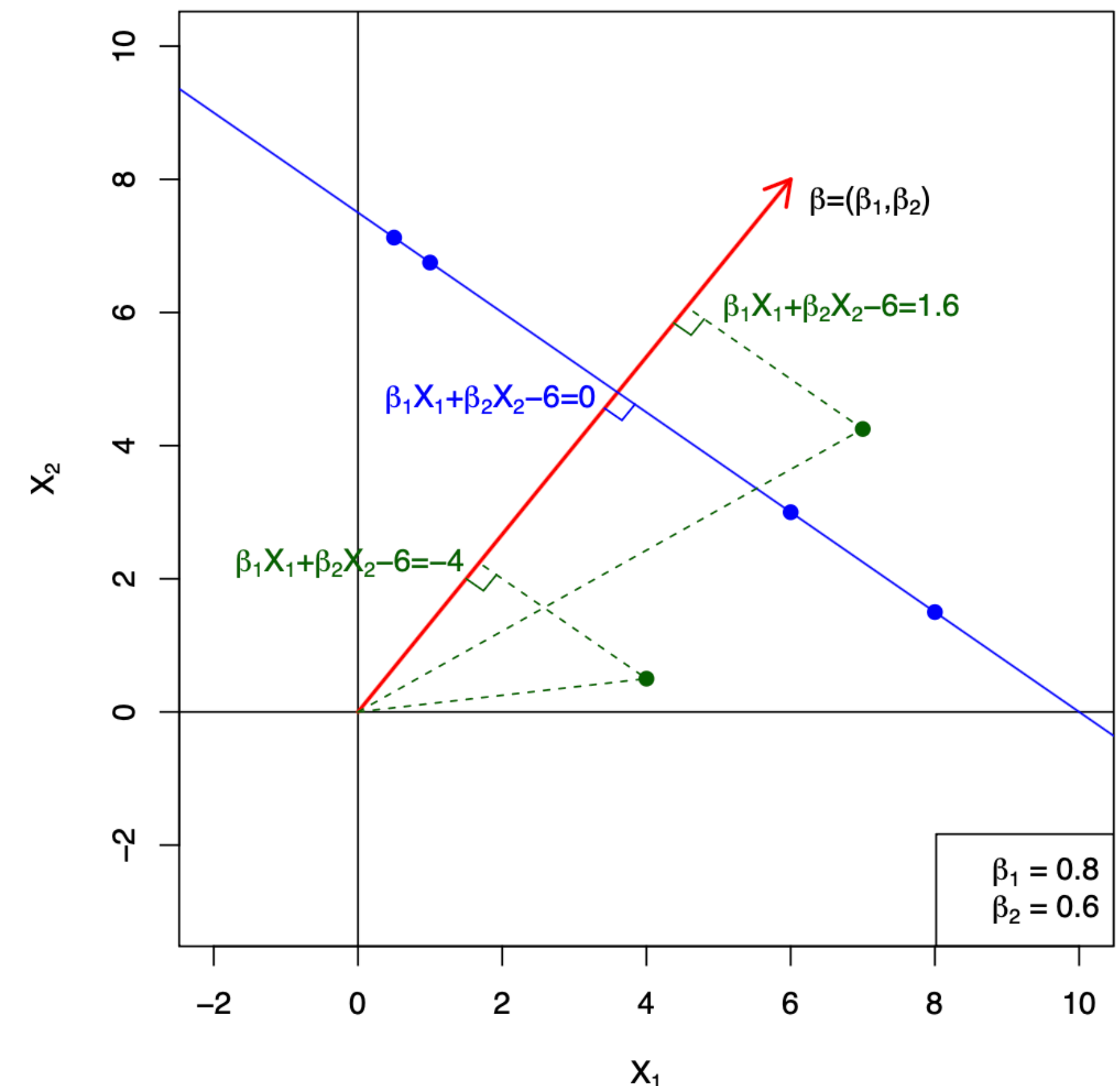
- Hyperplane in  $p$  dimensions is a flat affine subspace of dimension  $p - 1$
- General the equation for a hyperplane has the form

$$\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p = 0$$

- For  $p = 2$  we have a straight line

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$$

- The vector  $\boldsymbol{\beta} = (\beta_1, \beta_2, \dots, \beta_p)^T$  is called the **normal vector** - it points in a direction orthogonal to the surface of the hyperplane.

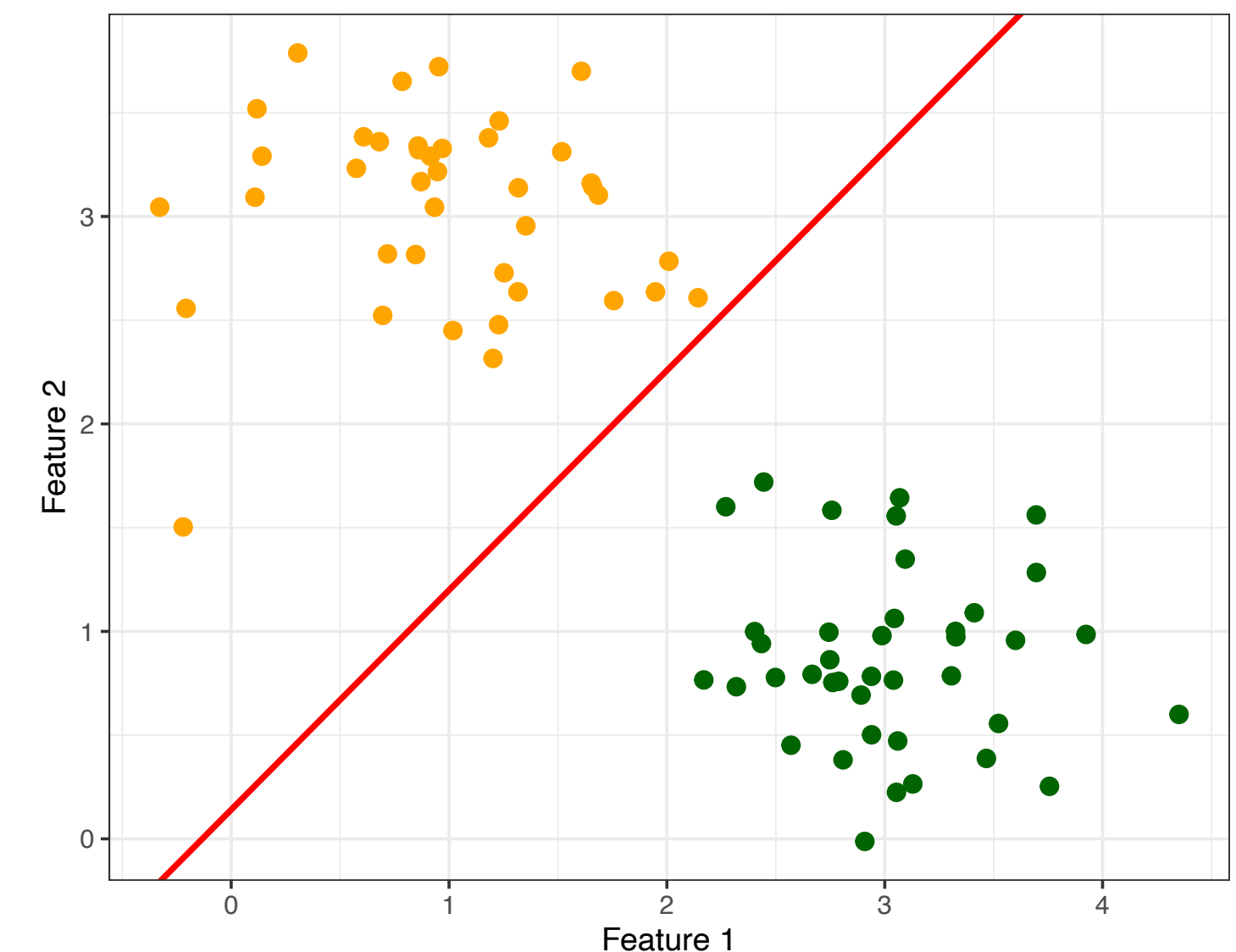
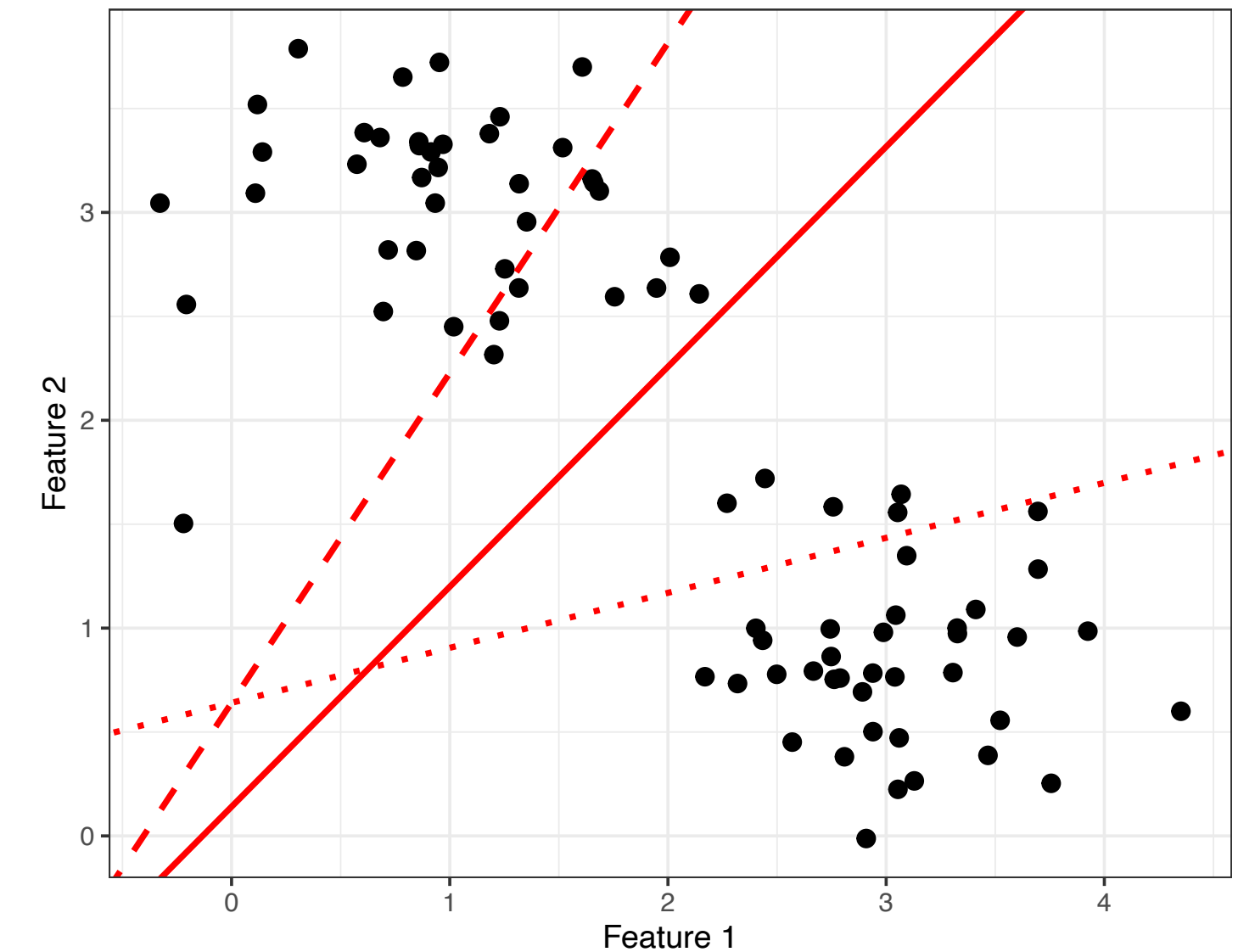


# Separating hyperplanes

$f(X)$  describes a hyperplane + visualizing for  $p = 2$  dimensions

$$f(X) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

- $f(X) > 0$  for points on one side of the hyperplane
- $f(X) < 0$  for points on other side.
- Given features  $X = (X_1, \dots, X_p)^T$ , assign **numeric** label ( $Y = \pm 1$ ) based on  $f(X) \lessgtr 0$
- If  $Y_i \cdot f(X_i) > 0$  for all  $i$ , then  $f(X) = 0$  describes a separating hyperplane.



# Maximal margin classifier (MMC)

MMC = hyperplane that provides strongest separation

Among all separating hyperplanes, find the one that makes the biggest gap or margin between the two classes.

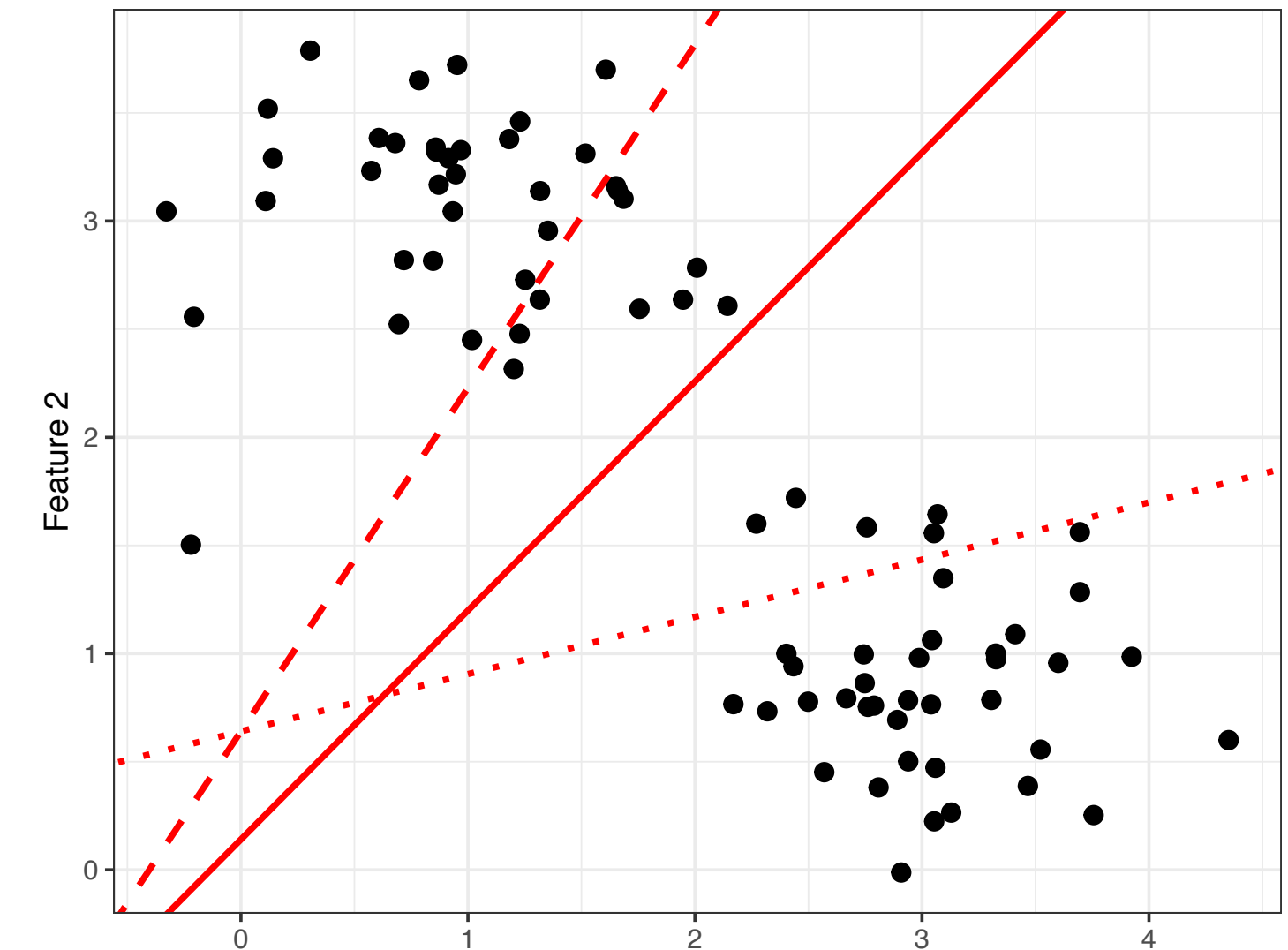
Maximize  $M$  with respect to  $\beta = (\beta_1, \dots, \beta_p)^T$  subject to

1.  $\sum_{j=1}^p \beta_j^2 = 1$
2.  $y_i(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}) \geq M$  for all  $i = 1, \dots, n$ .

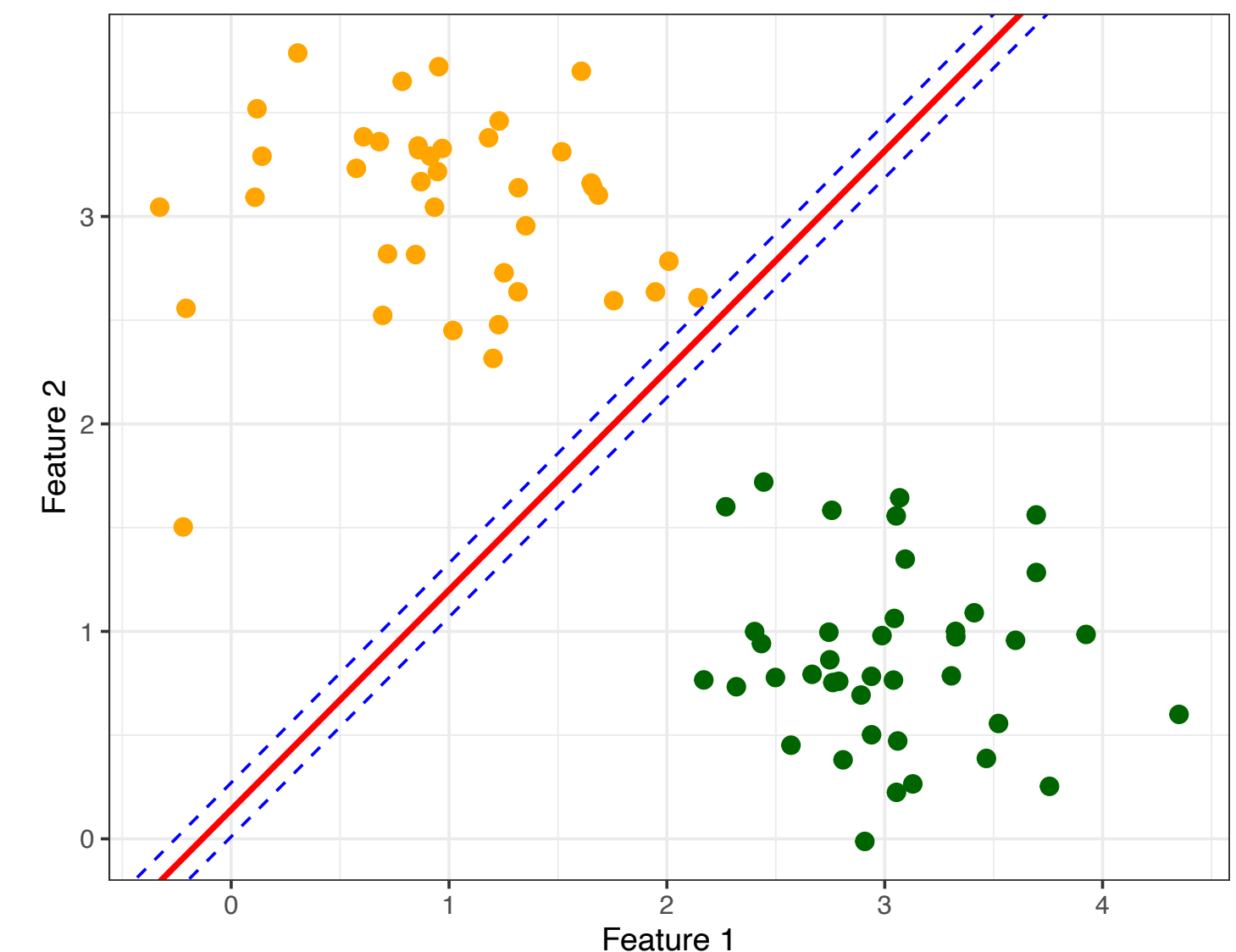
Constraint #2 defines a (candidate) hyperplane.

Constraint #1: distance from the  $i$ -th observation to the hyperplane is

$$y_i(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip})$$



$M$  = distance between red and blue lines



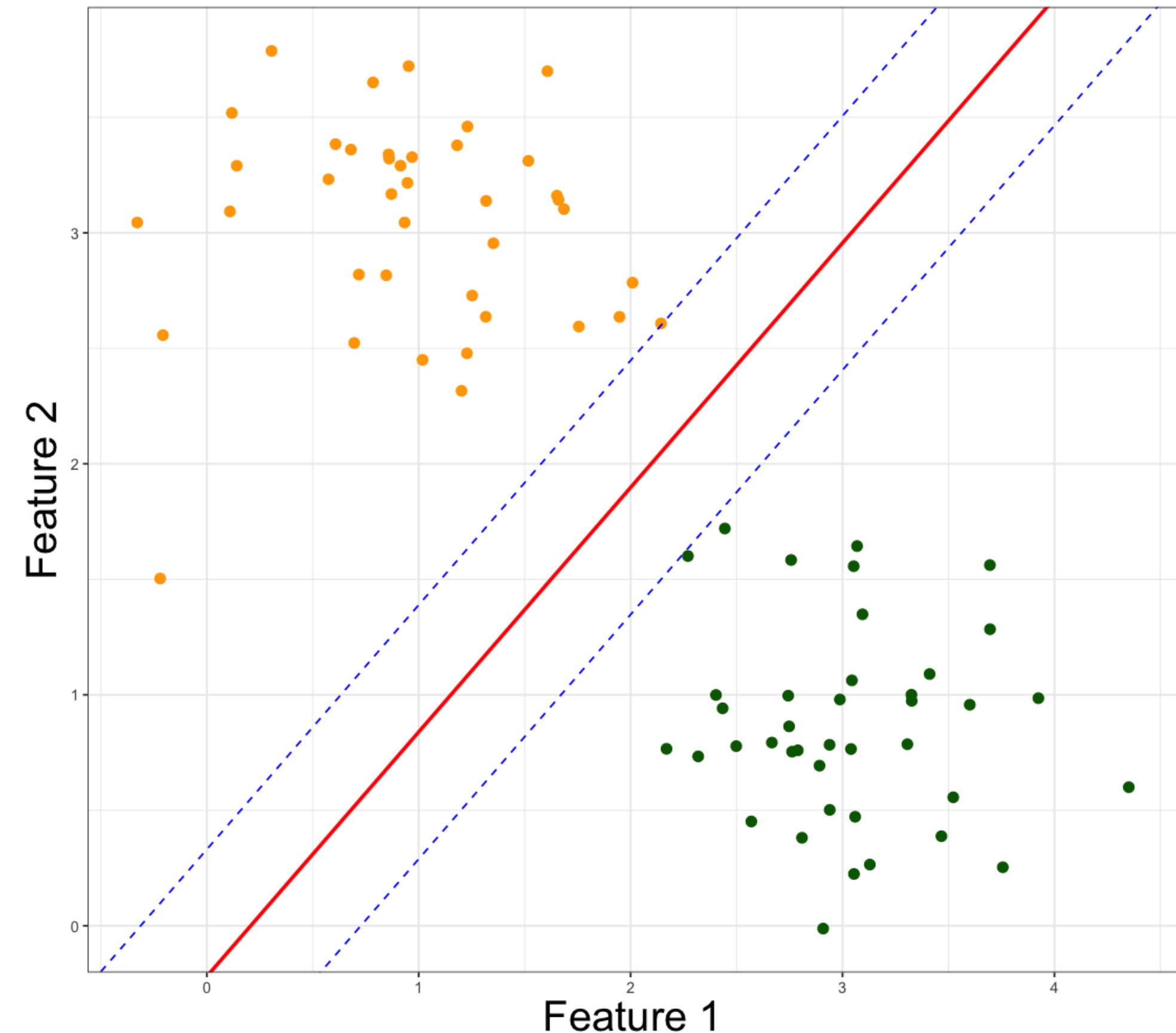
# Support Vectors

## Points on the edge of (dashed) margins

The observations that lie on the edge of the margin are called support vectors. These are the most critical data points; if you move them, the hyperplane may move.

Points further from the margin have no influence on the classifier.

**Key Limitation:** The MMC only works if the data is perfectly linearly separable. What happens in the real world when classes overlap?



# Support Vectors in the wild

## Linear Support Vector Classifier (Soft Margin)

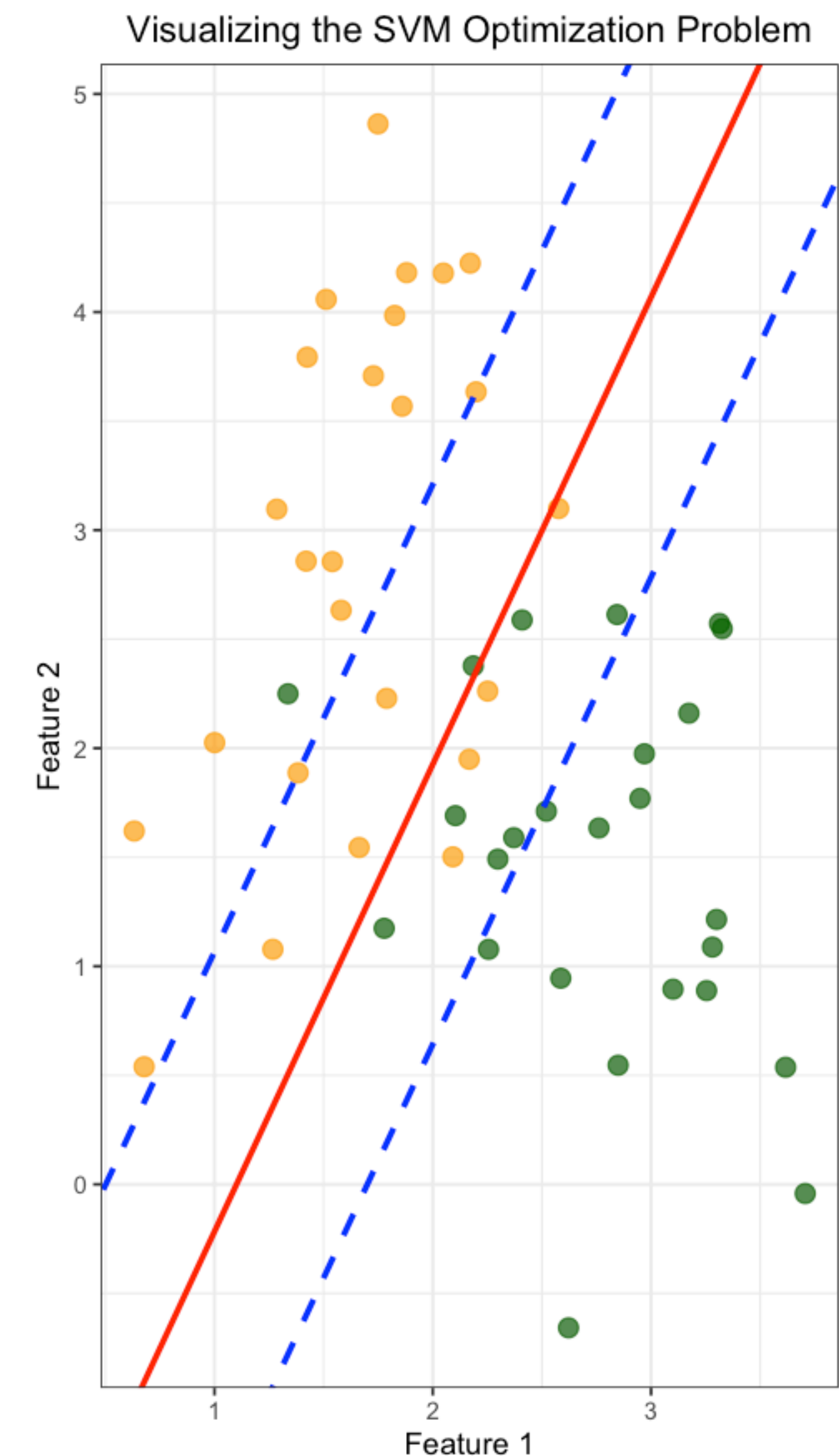
What if the classes overlap? No separating hyperplane exists!

*We can allow some observations to be on the wrong side of the margin (or even the hyperplane). This is a "soft margin". Introduces a trade-off:*

1. Maximize the margin width.
2. Limit the number of "margin violations"

Maximize  $M$  with respect to  $\beta = (\beta_1, \dots, \beta_p)^T$  and  $\epsilon = (\epsilon_1, \dots, \epsilon_n)^T$  subject to

1.  $\sum_{j=1}^p \beta_j^2 = 1$
2.  $y_i(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}) \geq M(1 - \epsilon_i)$  for all  $i = 1, \dots, n$ .
3.  $\epsilon_i \geq 0$  and  $\sum_{i=1}^n \epsilon_i \leq C$

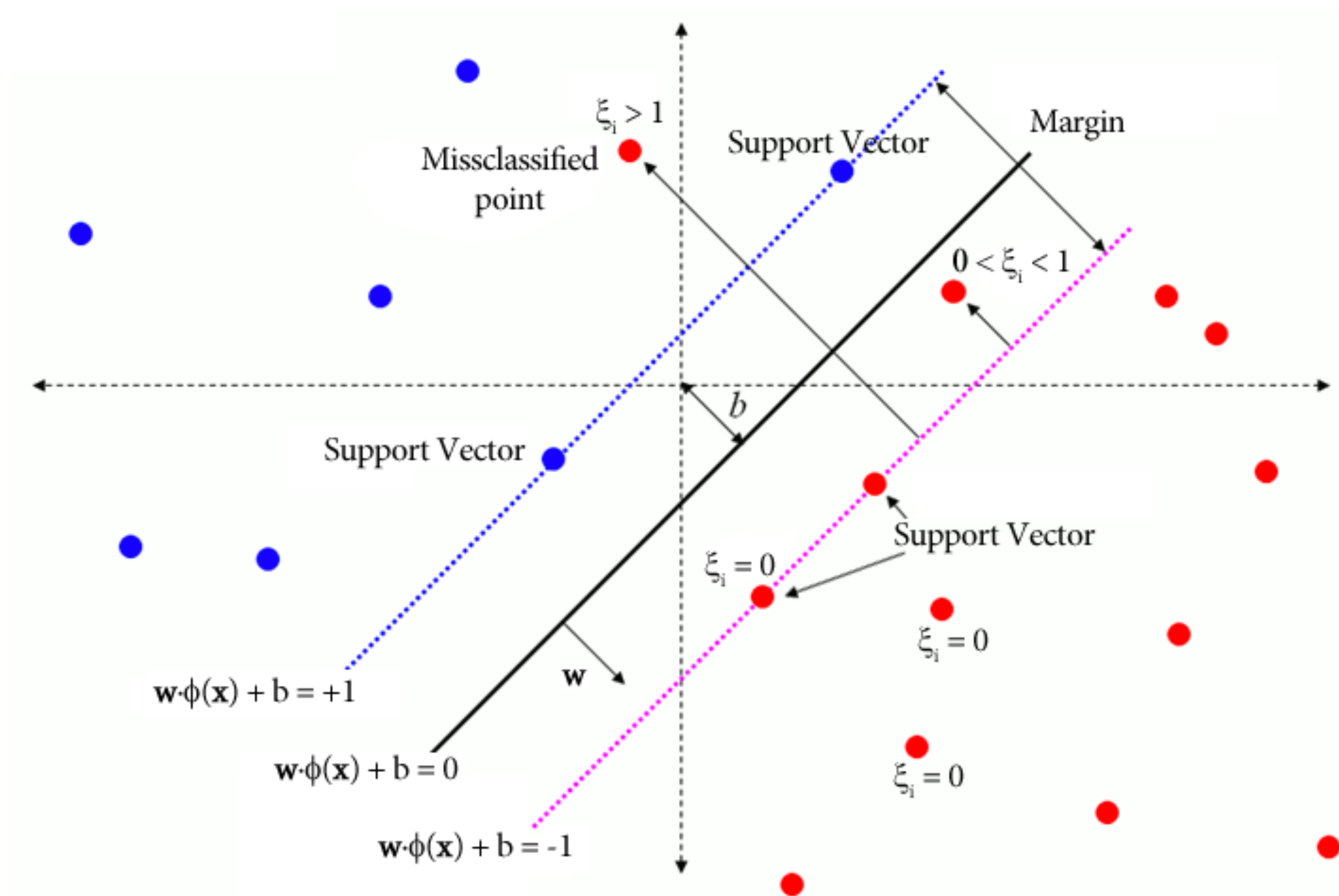


$\epsilon$  tells us where a point is located.

$\epsilon = 0$ : point is on correct side of margin

$\epsilon > 0$ : wrong side of margin

$\epsilon > 1$ : wrong side of hyperplane



# Support Vectors in the wild

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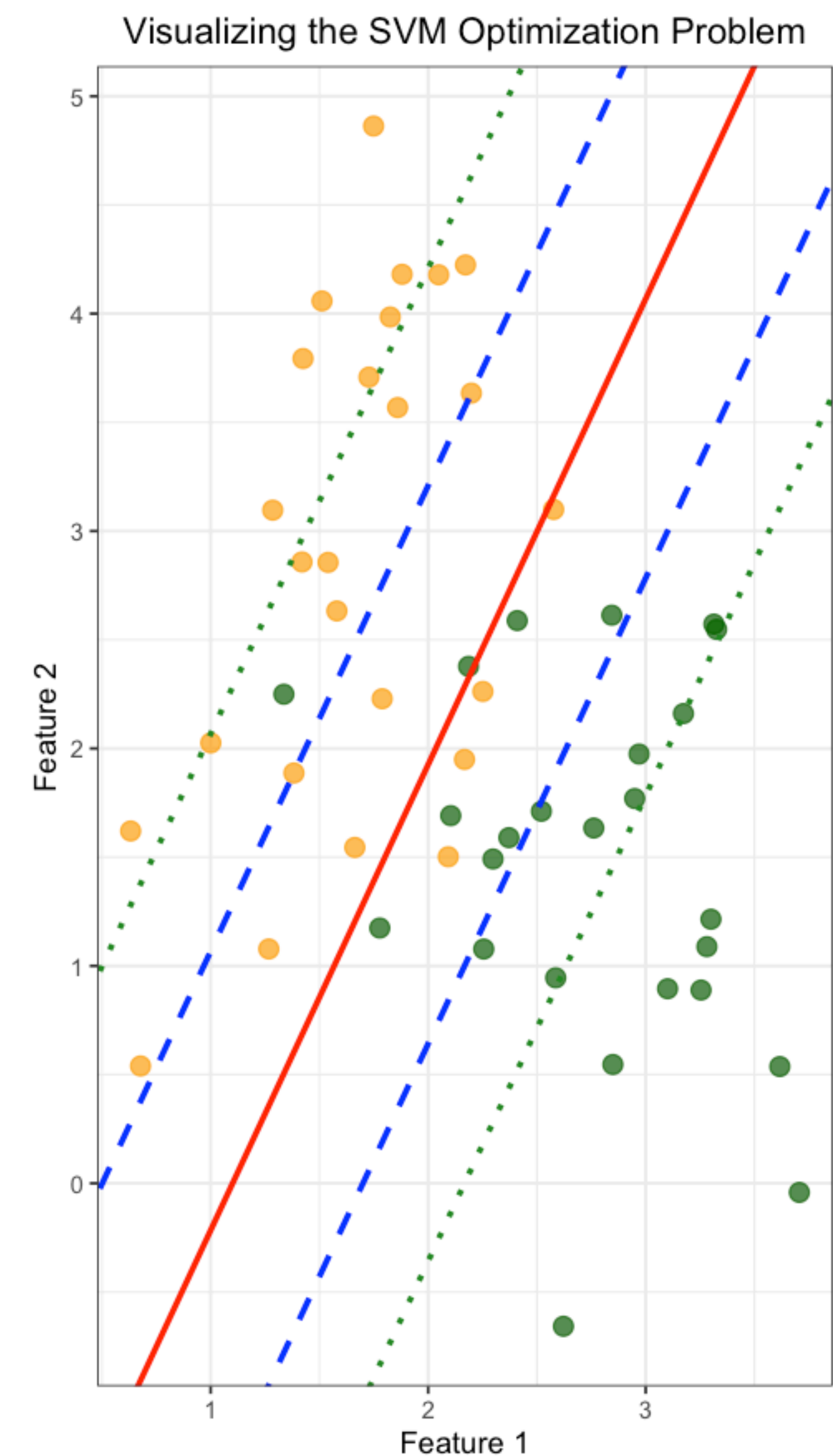
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How to choose  $C$ ?



**Large  $C$ :** high tolerance for observations being on the wrong side

**Small  $C$ :** little tolerance for observations being on the wrong side.

# Non-linear SVCs

## Enlarging feature space using transformations

Sometimes a linear boundary won't work!

*Enlarge the space of features by including transformations;*

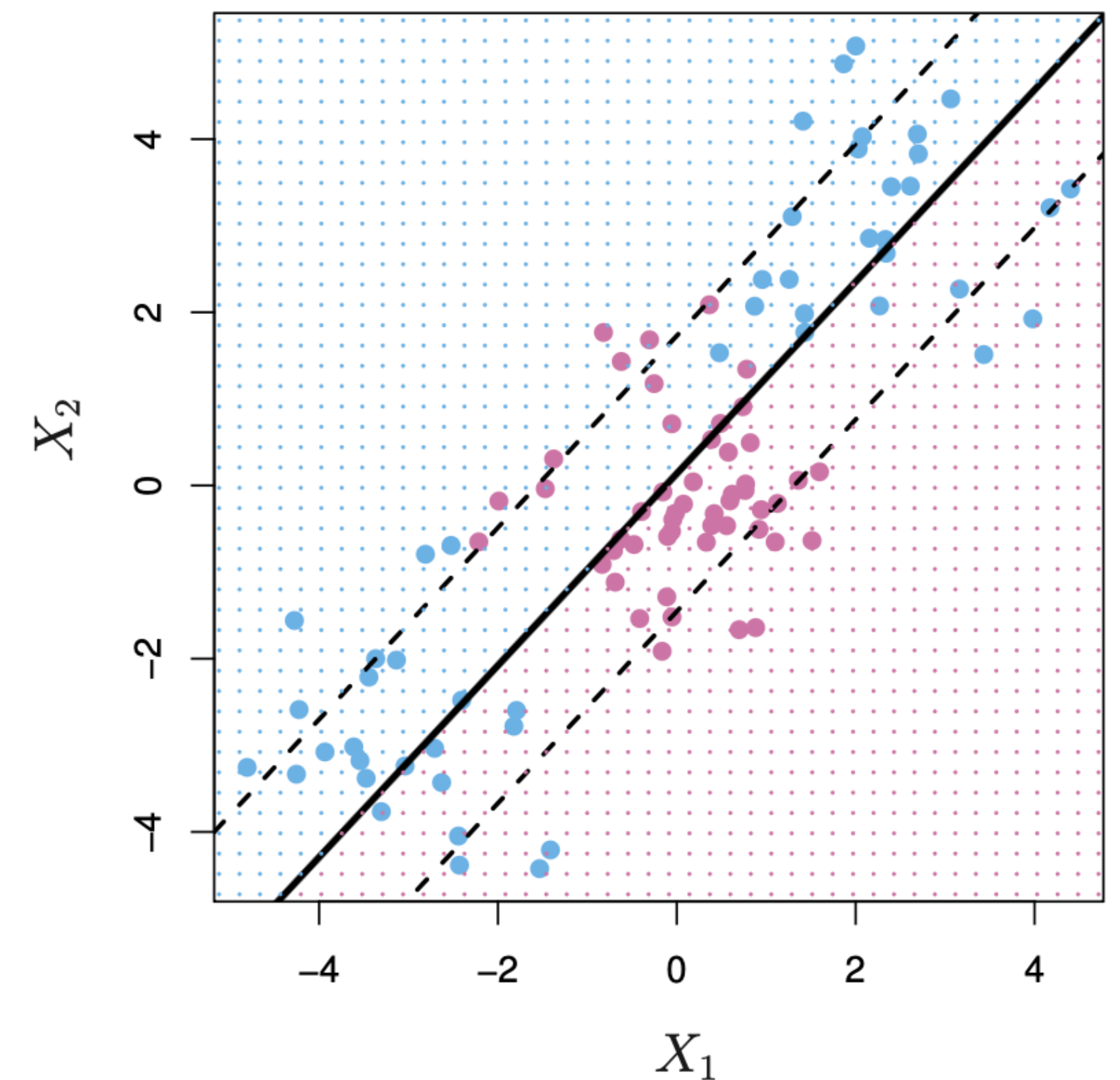
Example:  $X_1^2$ ,  $X_2^2$ ,  $X_1X_2$ , and  $X_1X_2^2$ .

*Go from a  $p$ -dimensional to an  $M > p$  dimensional feature space.  
Then, fit a support-vector classifier in the enlarged space.*

Results in non-linear decision boundaries in the original feature space.

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2 + \beta_6 X_1 X_2^2 \leq 0$$

Polynomial transformations scale very quickly:  
how many parameters for cubic polynomials?



# Tackling non-linearities using SVCs

Inner products and kernel functions 🤔

**Inner product** of two vectors  $\mathbf{x}_i$  and  $\mathbf{y}_i$ :  $\langle \mathbf{x}_i, \mathbf{y}_i \rangle = \sum_{j=1}^p x_{ij} y_{ij}$

Linear SVC:  $f(\mathbf{x}) = \beta_0 + \sum_{i=1}^n \alpha_i \langle \mathbf{x}, \mathbf{x}_i \rangle$ . #parameters = #samples + 1

Support Vector **Machines (SVMs)**: use kernel functions instead of inner products

$$f(\mathbf{x}) = \beta_0 + \sum_{i=1}^n \alpha_i K(\mathbf{x}, \mathbf{x}_i)$$

$$K(\mathbf{y}, \mathbf{x}_i) = \left( 1 + \sum_{j=1}^p y_j x_{ij} \right)^d$$

Polynomial kernel

$$K(\mathbf{y}, \mathbf{x}_i) = \exp \left( -\gamma \sum_{j=1}^p |y_j - x_{ij}| \right)$$

$\mathcal{L}_1$ -norm kernel

$$K(\mathbf{y}, \mathbf{x}_i) = \exp \left( -\gamma \sum_{j=1}^p (y_j - x_{ij})^2 \right)$$

Radial kernel

# SVMs: more than 2 classes?

## Two competing approaches

SVMs as defined work for  $K = 2$  classes. What about  $K > 2$  classes?

### OVA: One-versus-All

1. Fit  $K$  different 2-class SVM classifiers:

$$\hat{f}_k(\mathbf{x}) \text{ for } k = 1, \dots, K$$

$\hat{f}_k(\mathbf{x}^\star)$  classifies  $\mathbf{x}^\star$  as class  $k$  vs 'other'.

2. Classify  $\mathbf{x}^\star$  to the class  $k^\star$ , where  
 $k^\star = \operatorname{argmax}_k \hat{f}_k(\mathbf{x}^\star)$

Computationally lighter!

### OVO: One-versus-One

1. Fit all  $\binom{K}{2}$  pairwise classifiers:  
 $\hat{f}_{kl}(\mathbf{x})$  for  $1 \leq k < l \leq K$

2. Classify  $\mathbf{x}^\star$  to the class  $k^\star$  that wins most pairwise comparisons.

Pick this when  $K$  is small

# Post-class debrief

## Today we discussed...

The Core Idea: Find the Best Separating Hyperplane

Hyperplane: A line (in 2D), a plane (in 3D), or a flat subspace that separates classes.

Goal: Find the optimal hyperplane that creates the largest possible margin (gap) between the classes.

- **Maximal Margin Classifier:** Finds the hyperplane with the widest margin. Limitation: Only works for perfectly linearly separable data. The points on the margin's edge are support vectors.
- **Support Vector Classifier (Soft Margin):** Allows some data points to be on the wrong side of the margin (or even the hyperplane). The Cost parameter tunes the trade-off between margin width and the number of violations.
- **Support Vector Machine (Non-Linear):** Uses the kernel trick to create non-linear decision boundaries by implicitly mapping features to a higher-dimensional space where they become separable.

### SVMs for multi-class problems

- **One-vs-One (OVO):** Creates a classifier for every pair of classes.
- **One-vs-All (OVA):** Creates a classifier for each class against all the others.



*That's all Folks!*